TUTOR/MENTOR NETWORK PROJECT
SOCIAL/SPATIAL ANALYSIS OF NON-PROFIT COLLABORATION

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Abstract— By identifying connective links between participants and participating organizations; more effective communication and backbone support for 'Collective Impact' can be realized by providing information for organizations to quickly evaluate and identify partner organizations with differing resources leading to more coordinated and efficient problem solving.

Index Terms—Social Network Analysis, geospatial visualization, Tutor/Mentor Connection, collaboration,

I. INTRODUCTION

Daniel F. Bassill, Director of the Tutor/Mentor Connection, would like the analysis team to focus on the Social Network Analysis (SNA) categories of conference participants, in order to gauge which areas of social networking need to increase participation in the Tutor/Mentor Connection (T/MC) program(s).

The client would like the analysis team to compare participants and participation from the first conference (1994) to the current conference(s) in order to show how involvement has changed over time. It is important to note that the T/MC hosts (2) conferences per year, one in the spring (typically during the month of May), and another in November. A map series will be produced to illustrate the change in landscape as it relates to the changes in participation by SNA category.

Client would like analysis team to look for organizations who have had a decline/rise in participation (active/inactive? Why? Change in organization's mission?). Client would also like for analysis team to look into the possible creation of a platform in which event organizers can get immediate feedback concerning the participants at their designated events.

II. RELATED WORK

- Social Network analysis flow map and geospatial (world map) visualization using SNA codes and zip codes from provided Excel spreadsheets of conferences attendees
- Social Network analysis (radial tree graph) of May '09 and November '09 T/MC conferences, created by DePaul University student, Kalyani Misra, utilizing "SNA codes" from provided Excel spreadsheets of conference attendees

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- Social Network analysis (radial tree graph) of May '09 and November '09 T/MC conferences, created by DePaul University student, Kalyani Misra, utilizing" Organization Names" from provided Excel spreadsheets of conference attendees
- Radial Tree Graph visualizing participation at the May '09 and November '09 conferences, as well as colored nodes depicting the individuals who attended BOTH conferences
- Geospatial visualization of the percentage of people living in poverty (2000 Census data) of the various neighborhoods (zip codes) of the city of Chicago done in a diverging color scheme (map also contains symbols to represent the locations of T/MC programs)
- "Birth to Work Mentoring Strategy" flowchart with embedded hyperlinks to various resources found on the Tutor/Mentor Connection website
- "Skills and Talents Worksheet" flowchart with embedded hyperlinks to various resources found on the Tutor/Mentor Connection website
- Tutor/Mentor Connection strategy map- interactive map with hyperlinks to

III. STATISTICS OF DATA SET USED

When initially analyzing the data (Excel spreadsheets reflecting conference attendance from 1994 to the present), it was noted that there were over fifty SNA codes that were being utilized by the organization, which made the information less meaningful. Analysis team members then evaluated the SNA codes, and after further investigation, was able to reduce the number of distinct codes to thirty.

From 1994 to the present, there have been 4,563 individual participants at the Tutor/Mentor Connection conferences, with most of these participants being located in Chicago and its surrounding areas. The analysis team has asked the client for additional information concerning the specific location of each conference in order to determine if the geographical location of the actual conference has any effect on conference participation.

IV. DATA ANALYSIS/VISUALIZATION (ALGORITHMS) APPLIED

Analysis team members used Microsoft Excel to create a line graph (*Figure 5.1*) of conference attendance from 1994 to the present. The team also created Excel spreadsheets (digital/hard copies presented to the client) in order to acquire an accurate count of the number of times individuals participated in a Tutor/Mentor Connection conference.

The analysis team created a registration form via Google Docs, which could be shared via email, facebook, and Twitter, that would allow conference participants to identify themselves, using one of the new titles (Executive, Manager, Coordinator, Office Staff, Mentor, Volunteer, and Student) and SNA category. Once individuals have completed the check-in process prior to the start of the conference, their self-assigned title would be automatically reorganized into CartoDB, thus allowing for the creation of a geospatial visualization that would allow other participants to find desired title/SNA holders within their geographic location (*Figure 7.3*).

The analysis for *Figure 7.1* was performed on the V6 data set and utilized Gephi, and Inkscape. The grouping of attendance by organization yielded 1776 nodes; this made the map virtually unreadable, thus, the collection was filtered to those who attended four or more conferences, which resulted in the 167 nodes presented in the visualization.

Figure 4.1 (a – d) were created by downloading the zip code boundary files from the US Census Bureau Tiger Line website. Using QGIS, the Zip Code layer was intersected with the final version of data as described above. The point file of the attendees was processed using the tool: Points in Polygons (Vector > Analysis Tools > Points in Polygon). This resulted in a polygon layer which counted the number of points per polygon. Maps 4 c and d were processed to illustrate the senior leadership (Executive and Managers) participation in Tutor/Mentor Network. Program and non-program senior leadership participation tends to have a NE Illinois center of gravity. The dataset was then saved to GeoJSON file format and uploaded to the online mapping system CartoDB (Figure 4e). This tool was chosen to display the geographic information as it provides a platform for sharing the information. The low-cost (free) account can easily be used to display and share gathered geographic information.

V. Key Insights Gained (Initial Analysis):

When examining Tutor/Mentor Connection conference attendance, the analysis team wanted to further investigate the rationale behind why some conferences were more heavily attended than others. The analysis team came to the conclusion that May '99 was the most attended conference with 350 participants. The May conferences generally, though not always, have higher attendance than the November conference of the same year. The seasonality of attendance may be due to an end-of-school-year effect, whereby there is a higher perception of need due to the temporal proximity to the conclusion of a full term of schooling and grading. A closer analysis of the attendee type (SNA Category and/or Organization) and count, and season of the conferences through a scatter plot would show this correlation (or disprove the theory). The plot of attendance shows some similarity to the typical "hype curve". This trend would suggest is that the conferences have matured to the point where attendance levels and participation are at a highly effective level: the so-called "Plateau of Productivity".

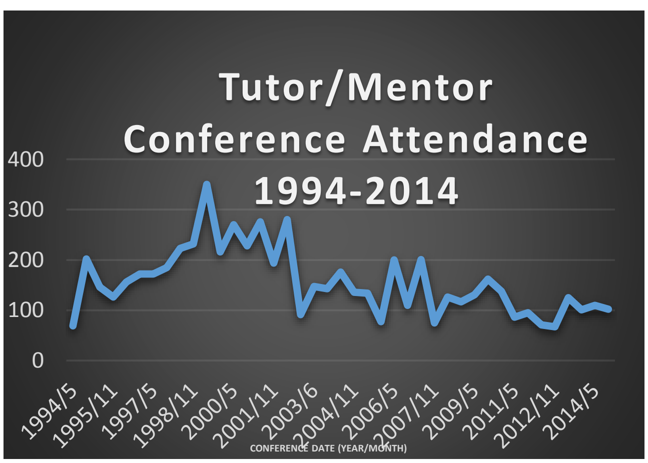

Figure 5.1

VI. Validation Issues/Re-Design Resolution

The analysis team was faced with the issue of ensuring that we were able to provide the client with visualizations that were meaningful and "told a story". One of the major issues the team faced was the enormous amount of data that needed to be "cleaned". From the Excel spreadsheets provided by the client, the team was able to reduce the number of Unique Organization values from 2445 to 1775. In order to maintain a reference to the

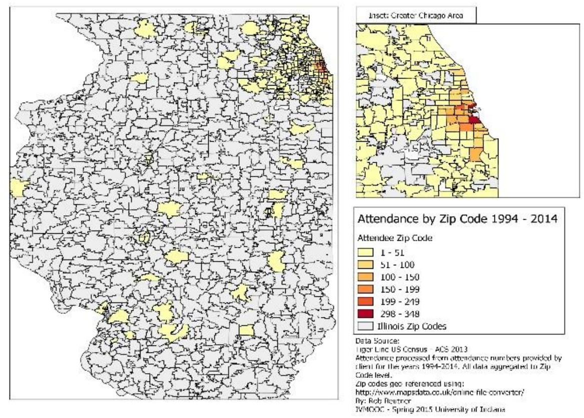

Figure 4.1(a) – Zip Codes with the highest rates of attendance

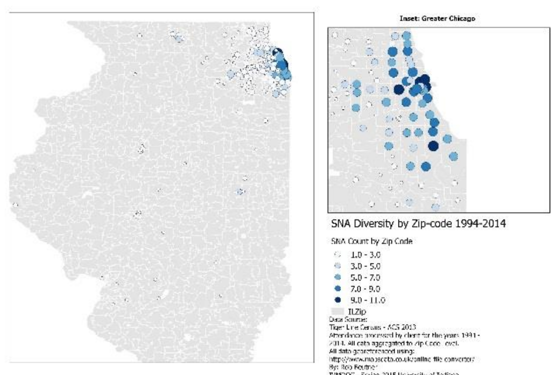

Figure 4(b) – Diversity index of SNA codes by Zip Code

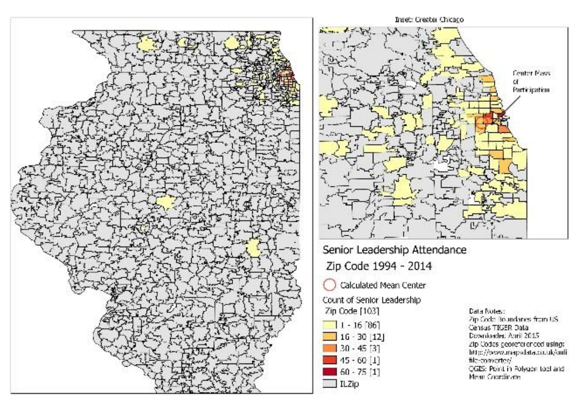

Figure 4 (c) – Attendance at conferences by Program Senior Leadership for 1994- 2014. Note the center of gravity for the attendance is located in SW Greater Chicago.

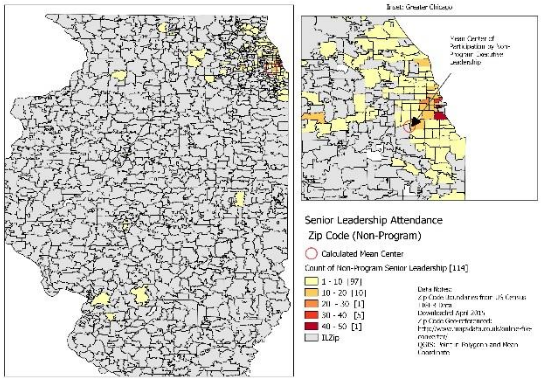

Figure 4(d) - Attendance at conferences by Non-Program Senior Leadership for 1994- 2014. Note the center of gravity for the attendance is located in SW Greater Chicago. Non-program participation appears to reach further south in Illinois as opposed to a more restricted distribution of Program participation.

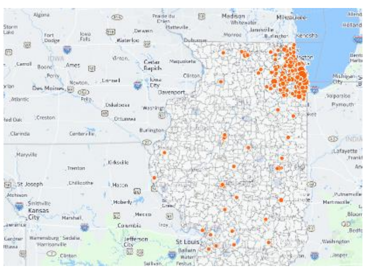

Figure 4(e) – SNA diversity. Interactive map.
http://cdb.io/1GwAFm2

original values from the client, the analysis team kept two columns for *SNA Category* and two columns for *Organization,* within the context of the revised client spreadsheets, as follows:

- SNA Category - ORIGINAL
- SNA Category - CLEANED
- Organization - ORIGINAL
- Organization – CLEANED

In continuing with this trend, the team also cleaned the Titles category, which reduced the number of unique identification labels from 1499 to 437. While this reduction in labels would aid in the visualization process, it was decided that further "cleaning" would enhance the data. The analysis team decided to create "buckets" for the title category, which would allow for a more streamlined visualization. The newly created buckets were as follows:

- Executive
- Manager
- Coordinator
- Office Staff
- Mentor
- Volunteer
- Student

The analysis team then created a new column and used a VLOOKUP to pull the new assigned value into the main tab and then copied and pasted the values in another column, thus giving the team two new columns, which were "Assigned Title with VLOOKUP" and "Assigned Title Hardcoded". The VLOOKUP column was allowed to remain, in case some values need to be reassigned from the other tab.

In order to further the investigative process of the analysis team, the client was asked to revisit the collected data in order to determine if organizations/programs were still active participants in the Tutor/Mentor Connection conferences. Daniel Bassill, client, placed an "Active/Inactive" label on all of the organizations in the "Programs" category, using the best information available. For instance, if he could not find them in a Google search, and had no recent contact or awareness of them, he coded them "inactive". It should be noted that being coded "inactive" does not mean the organization no longer exists, it could simply mean that there is no evidence on a web site, demonstrating that this particular organization is currently operating an organized tutor/mentor program.

While researching the activity of each organization, the client was able to re-connect with people who had attended T/MC conferences in the past. Daniel has also started a Facebook list of links to organizations who have previously attended. The client will continue to try to encourage connections with the T/MC, and each other, as part of the overall effort to build collaboration.

VII. Visualization Stories

When analyzing data, it is the expectation that a "story" emerges and can be communicated from the material that has been gathered. This particular visualization (Figure 7.1) was able to illustrate the history and trends in participation at the Tutor/Mentor Connection Conferences over the past two decades. The map depicts organizations as nodes, and each is related only to the Tutor Mentor Connection (the conference organizer). The size of the node illustrates the number of times an organization has attended: the larger the node the more times they have attended, the smaller the node the less frequent. The color of the node indicates how recently an organization has attended: the lighter in color (in the yellow spectrum) the less recent, and the darker the color (in the darker blue) the more recent. The organization affiliation is distinguished by the color of the node label: program in green, non-program in pink.

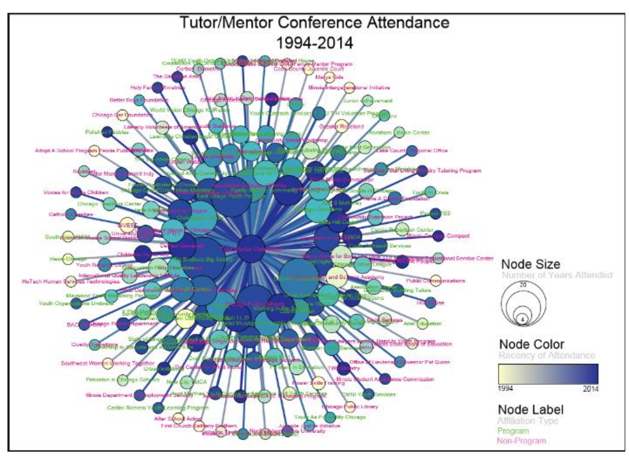

Figure 7.1

It is the hope of the analysis team that this visualization articulates to the client the organizations attending T/MC conferences, how often these organizations are attending, how recently have these organizations attended, if there are organizations who have come offering help (are they still attending?), and are there organizations who are seeking help.

Being able to answer these questions may afford the client the opportunity to follow-up with participants in order to discern whether or not conference participation allowed these organizations to meet their desired needs. Are effective and valuable relationships being built? Or, if an organization has stopped attending, the follow-up might inquire into the reasons why. Perhaps additional help is needed, or perhaps different categories of help are needed. Answers to these might drive the campaigning of organizations to attend.

In constructing the conference registration form (Figure 7.2), it is the hopes of the analysis team that the client, Daniel Bassill, will be able to analyze conference attendance based on geographic location (Figure 7.3). Ideally, the client would like to see participation from all areas of Chicago, with the hope of becoming an influential program, recognized at the national level. By allowing participants to geo-locate their organization, it allows for conference attendees, as well as organizers to understand the areas of Chicago (and possibly nation-wide) where conference attendance needs to be further stimulated. This new registration form also allows the attendees and client to see the various organizations/titles that are attending conferences, thus allowing for an opportunity for more valuable social networking, based on individual program needs.

Figure 7.2 (Left)

Figure 7.3 (Above)

VIII. Continuation of Collaboration

The analysis team will continue to collaborate with Daniel Bassill, client, in order to ensure his success in utilizing the proposed visualizations and programs. Team members will conduct future meetings with the client and will possibly participate in client hosted webinars in order to inform other organizations of the potential influence of these visualizations/tools.

Acknowledgments

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